

HEAD-MOUNTED VARIABLE-SPECTRUM PHOTOTHERAPY APPARATUS AND METHOD FOR AUTOMATED CIRCADIAN ALIGNMENT

Open-Source Technical Specifications & Prior Art Architecture

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1. FIELD OF THE INVENTION

This disclosure relates generally to head-mounted therapeutic lighting systems, and more specifically to an open-source, wearable, variable-spectrum phototherapy apparatus integrated directly into cold-weather headgear and an unnetworked, localized algorithmic method for providing distinct, non-continuous circadian anchors to simulate a natural diurnal solar cycle in high-latitude, low-solar, or polar environments.

2. BACKGROUND & PROBLEM STATEMENT

Human physiological systems are profoundly regulated by the circadian master clock, situated within the suprachiasmatic nucleus (SCN) of the hypothalamus. This master clock relies heavily on external photic cues to synchronize the body's internal 24-hour cycle with the external environment, regulating critical endocrinological processes including the suppression and secretion of melatonin and the timed release of cortisol.

In high-latitude regions, geographic areas undergoing extended seasonal polar nights, or entirely enclosed artificial environments, the complete absence of natural sunlight leads to severe circadian misalignment. This desynchronization frequently manifests as seasonal affective disorder (SAD), profound fatigue, insomnia, cognitive degradation, and fractured sleep architecture. Existing light-therapy systems, such as stationary high-lux desktop lightboxes, suffer from extremely low user compliance due to their restrictive stationary requirements. Alternative wearable solutions, such as optical glass frames, feature rigid and delicate structures designed exclusively for stable indoor use, rendering them entirely incompatible with the rigorous mechanical and thermal insulation requirements of protective outdoor cold-weather apparel.

3. STRUCTURAL ARCHITECTURE & PHYSICAL GEOMETRY

The present invention establishes a comprehensive framework that integrates functional phototherapy architecture directly into the textile array of wearable headgear (e.g., a brimmed cap, a flexible knit winter beanie, or an insulated hood). This configuration ensures constant biological optimization while protecting the electronic architecture from extreme ambient cold and moisture.

The device includes an optical assembly embedded flatly into the underside of the headgear brim, positioned directly above the wearer's upper eye frame at a proximal distance threshold of less than 5 centimeters from the cornea. This optical layer comprises an edge-lit light guide plate paired structurally with a high-efficiency diffusion layer. Microscopic solid-state light sources are arranged along the lateral edges of the light guide plate, projecting

light inward into the panel. The light guide plate and diffuser redirect this light uniformly downward, translating sharp points of light into an ambient, non-glare volumetric wash that bathes the user’s ocular frame. By situating the light source overhead, the generated photons strike the lower and nasal regions of the retina. This configuration maximizes structural alignment with the biological distribution of intrinsically photosensitive retinal ganglion cells (*ipRGCs*), which are hardwired to receive downwelling light from the natural sky to anchor circadian rhythms.

4. ELECTRONIC FRAMEWORK & LOGICAL FLOW

To eliminate reliance on external networks, global positioning satellites (GPS), or auxiliary smart devices—which frequently fail or drop connectivity in isolated high-latitude zones—the apparatus utilizes a completely localized, self-contained electronic hardware layout. This structure is detailed broadly in the schematic block diagram below.

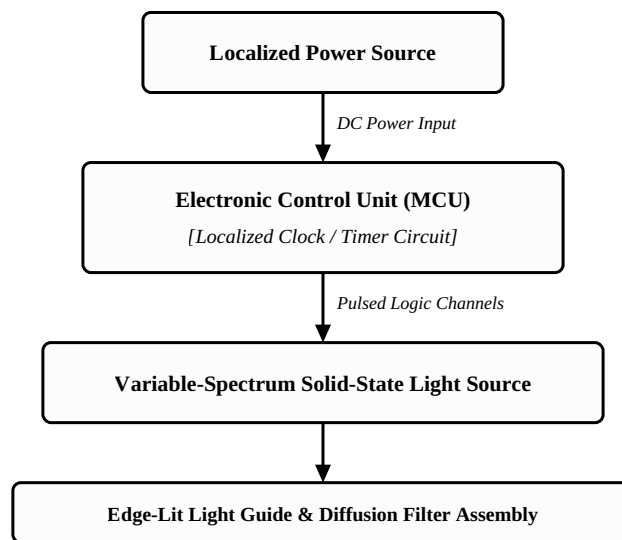


Figure 1: Schematic layout of the localized, broad functional architecture blocks.

The components of this system are functionally defined as follows:

- **Localized Electrical Power Source:** A self-contained energy cell network structurally embedded within the headgear (e.g., inside the rear adjustments, inner linings, or brim seams). This covers all electrochemical storage mechanisms, including but not limited to rechargeable or disposable configurations of any chemical composition (including lithium-ion, lithium-polymer, lithium-iron-phosphate, alkaline, nickel-metal hydride, or iron-based architectures), or direct tethered cable inputs from secondary external storage packs.
- **Electronic Control Unit:** An integrated processing circuit, microcontroller, or logic controller executing firmware routines tied entirely to a localized internal hardware clock or timer circuit. The system operates autonomously without needing external network validation, cellular signals, or satellite timing sync.
- **Variable-Spectrum Solid-State Light Source:** A multi-channel emitting array (such as multi-die LEDs, organic light-emitting structures, or micro-LED arrangements) containing distinct, selectively-driven spectral emission nodes. These channels specifically cover a short-wavelength circadian excitation band (peaking heavily between **460–480 nm**) and a monochromatic long-wavelength circadian-blind sleep preparation band (peaking at **660 nm**).

- **Edge-Lit Light Guide & Diffusion Filter Assembly:** A light transmission panel mounted flatly beneath the brim structure that captures raw, localized high-intensity emissions and distributes them smoothly across the user's lower retinal region as a low-intensity, homogeneous wash.

5. THE MULTI-STAGE CIRCADIAN SHIFTING ALGORITHM

To avoid user visual fatigue and ensure optimal energy efficiency, the electronic control unit is configured to deliver a pulsed diurnal cycle. The *ipRGC* photoreceptors require continuous exposure blocks to achieve full kinetic phototransduction signaling thresholds due to their slow integration kinetics. Brief, fractured flashes (e.g., one minute every ten minutes) fail to reach this therapeutic dosage, creating conflicting clock signals and causing severe visual orientation distraction.

The system resolves this by automating five distinct, non-continuous phototherapy exposure sessions distributed systematically across a 12-hour waking timeline. Each session runs for an uninterrupted duration of approximately 30 to 45 minutes, satisfying receptor integration kinetics while conserving battery life. The automated sequence is defined in the profile mapping below:

SESSION ID	TARGET TIMELINE	DOMINANT SPECTRUM TARGET	OCULAR ILLUMINANCE & POWER SPECS	BIOLOGICAL INTENTION & HORMONAL TARGET
01: Wake Anchor	07:00 – 07:45	High-Color-Temp Cool White (~6500K) Peak emission: 460–480 nm	~200 – 250 Lux at cornea Total output: ~30 – 50 Lumens	Suppresses melatonin completely; spikes morning cortisol; locks in 24-hour master clock initialization.
02: Morning Stabilizer	10:30 – 11:15	Sustained Sky Blue / Cyan Mix (~5000K) Broad short-wavelength band	~150 – 200 Lux at cornea Total output: ~25 – 40 Lumens	Reinforces daytime internal tracking; prevents phase-shifting clock drift or "free-running" states.
03: Slump Buster	14:00 – 14:45	Crisp Balanced Daylight White (~5500K) Moderate blue concentration	~100 – 150 Lux at cornea Total output: ~20 – 30 Lumens	Counters the natural postprandial dip in metabolic alertness; optimizes afternoon cognitive capacity.
04: Transition Anchor	17:00 – 17:45	Warm Amber / Low-Kelvin Gold (~2700K – 3000K) Short-wavelength blue omitted	~30 – 50 Lux at cornea Total output: ~5 – 10 Lumens	Signals daytime termination; allows gradual initiation of endogenous melatonin production pipelines.
05: Sunset Lock	20:00 – 20:45	Pure Monochromatic Deep Red Narrowband emission: 660 nm	< 10 Lux at cornea Total output: ~1 – 2 Lumens	Provides visual orientation lighting without triggering circadian receptors; preserves pre-sleep melatonin accumulation.

By enforcing this structured interval framework, the total operational time of the active light matrix is restricted to less than 4 hours per day. Operating at an optimized low-intensity layout of less than 0.5 Watts at peak output, the aggregate daily consumption is confined to less than 2 Watt-hours ($\leq 2 \text{ Wh}$). This allows standard compact power modules to drive the device reliably for multiple days without requiring physical re-charging infrastructure.

6. OPEN-SOURCE STATUS & PRIOR ART DEFENSIVE SCOPE

This technical specification documentation is explicitly published to the public domain as defensive prior art to guarantee open access and prevent any third-party entities from capturing the core architectural logic under restrictive utility patent legal frameworks. Under the conditions of the GNU General Public License v3.0 (GNU GPL v3.0), any future physical implementations, schematic variations, derivative electronic hardware design iterations, or customized firmware forks must remain entirely transparent, open-source, and unencumbered by proprietary restrictions. The defensive prior art boundary established by this documentation explicitly includes:

1. Any wearable clothing item, protective gear, or headgear incorporating an interior or downward-facing lighting matrix configured to illuminate an individual's ocular field for biological or circadian clock regulation.
2. The use of non-continuous, block-interval phototherapy pulses (specifically ranging from 20 to 60 minutes) separated by distinct multi-hour blocks of inactive dormancy to simulate a full diurnal cycle.
3. Any implementation of multi-stage spectrum shifting using a pre-programmed standalone internal microcontroller that alters light composition across distinct steps, including cool blue-white, sky blue, daylight white, amber, and narrowband red wavelengths, without requiring external network dependencies.
4. Any optical geometric alignment wherein a wearable visor or brim utilizes light guides or diffusers to focus light specifically downward into the lower region of the human retina to target circadian-active ganglionic cell layers.